

Chapter 1

Markov Chain

1.1 Discrete Parameter Markov Chain

1.1.1 Introduction

Definition 1.1.1 (Stochastic Process). A stochastic process $\mathbf{X} = \{X(t), t \in \mathcal{T}\}$ is a collection of random variables. That is, for each t in the index set $\mathcal{T}$, $X(t)$ is a random variable.

The index is often interpreted as *time* and, as a result, we refer to $X(t)$ as the *state* of the process at time t . The *state space* of a stochastic process is defined as the set of all possible values that the random variables $X(t)$ can assume. For instance,

- $\{X(t), t = 0, 1, \dots\}$ is a discrete-time stochastic process indexed by the nonnegative integers;
- $\{X(t), t \geq 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

Thus, a stochastic process is a family of random variables that describes the system evolution through time of some (physical) process. This gives the conceptual fundation for *Time-Dependent System Reliability*.

Markov Property states that given present state, forecast of future is independent of past, i.e.,

$$P\{\text{Futhure} \mid \text{Present, Past}\} = P\{\text{Futhure} \mid \text{Present}\}.$$

Then, a stochastic process that follows the Markov property is called a *Markov process*.

Definition 1.1.2 (Discrete-Parameter Markov Chain). Let $\{X_n, n = 0, 1, \dots\}$ be a stochastic process that takes on a finite or countable number of possible values. Suppose that

$$P\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0\} = P\{X_{n+1} = j \mid X_n = i\}$$

for all states $i_0, i_i, \dots, i, j$ and all $n \geq 0$, then stochastic process $\{X_n, n = 0, 1, \dots\}$

is known as a discrete-parameter Markov chain (DPMC). In addition, if $P\{X_{n+1} = j | X_n = i\} = p_{ij}$ for any $n \geq 0$, it is a homogeneous DPMC.

In homogeneous DPMC, p_{ij} represents the probability that the process will when in state i , next make a transition into state j , which is call one-step transition probability. Let $\mathbf{P}$ denote the matrix of p_{ij} , so that the *transition probability matrix* is

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ p_{i0} & p_{i1} & p_{i2} & \cdots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix},$$

where $0 \leq p_{ij} \leq 1$ and $\sum_j p_{ij} = 1$ for all i .